

COMPARISON OF BRUECKNER THEORY WITH "EXACT" RESULTS FOR H^3 AND He^4 NUCLEI

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The consistency of the second-order Brueckner-Baranger approximation to the ground-state energy E of H^3 and He^4 nuclei is confirmed by the agreement of E with the recent Faddeev and Faddeev-Yakubovsky equation results for Reid local potential, by the dependence of E on different phase-equivalent transforms (= off-shell changes) of the Reid potential which agrees with Faddeev equation and nuclear matter calculations, and by the Coester line well reproduced for a set of realistic potentials in He^4 . The influence of approximations in solving Bethe-Goldstone equation is also discussed.

1. INTRODUCTION

It is a paradox that the perturbation theory is often used with good results even without the formal convergence proof. The Brueckner theory of finite nuclei is a typical approximation technique of this type (cf. e.g. the review paper by Baranger [1]). This theory is closely related to the variational $\exp S$ formalism (Kümmel et al. [2]). In the lowest orders, the Brueckner theory (BT) preserves its virtue of being relatively simple perturbation approach without making extreme demands on the computer capacity.

In the present paper, we test this BT theory by comparison with "exact" results. In this aspect, we are continuing the work of Sotona [3]. Following consequently Baranger in his careful treatment of methodology of the lowest-order approximations (for a short description see section two), we employ for comparison the recent H^3 and He^4 energies obtained respectively by Kim et al. [4] and Tjon [5] from the specialized Faddeev and Faddeev-Yakubovsky integral equations. The technical details are summarized in section three and the results are presented and discussed in section four.

2. SURVEY OF THE METHOD

Splitting of the exact Schrödinger equation Hamiltonian H into the sum of a diagonal operator H_0 and a small perturbation V seems to fail in the nuclear theory context. This is replaced by using large "perturbations" (e.g. realistic potentials with a core) plus the Brueckner rearrangement technique (for details see [6]). Due to this concept the procedure of estimating diagrams must be done very carefully. For the needs

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of this introductory discussion, let us recall the example of the low order separable potential calculation of He^4 by Afnan et al. [7]. We may conclude from their results that the third order corrections to the radius indicate a slow convergence rate. In the region where the second-order energy contributions are minimized by the choice of H_0 (in fact, by the choice of the oscillator frequency Ω in the linear harmonic oscillator (LHO) Hamiltonian $H_0 = H_\Omega$), we observe that big changes in the radius are caused by different choices of the almost free parameter Ω . Hence we believe that even the third order is still insufficient and BT cannot replace the necessary and complicated variational calculations when the higher precision is required.

The similar situation is met in the ground-state energy evaluation. Fortunately, in contrast with the case of radius, the dependence of E on Ω is not monotonic and the smallness of the third-order contributions [8, 9] seems to support the applicability of BT (for this single quantity only!). Hence, we shall test the use of the Baranger's prescription for the determination of $\Omega = \Omega^{(B)}$ and E in the present paper. In order to avoid the possibility of any sort of fitting E by Ω , we shall employ the whole classes of the phase-equivalent potentials (cf. Kim et al., [4]) to obtain the reasonable and significant test of precision.

There exists a popular improvement of H_0 by shifting its spectrum in the desirable direction, $\varepsilon \rightarrow \varepsilon - C$. The parameter-independence arguments similar to the choice of Ω (after all, we should not fit or simulate the higher orders) seem to support the choice $C = 0$ advocated from the different point of view by Zabolitzky [10]. As a consequence, we are not allowed to use C for "compensation" of any diagram and pay for $C = 0$ by technical complications: The definition of the reaction matrix G (Bethe-Goldstone, BG equation) should read

$$(1) \quad G = V - V\mathfrak{A}G, \quad \mathfrak{A} = \mathfrak{A}(\omega) = Q(\omega + QTQ)^{-1}.$$

When compared with the standard propagator $Q(\omega + H_0)^{-1}$, the more involved structure of $\mathfrak{A}$ containing the kinetic energy operator T together with the Pauli projector Q in the denominator makes the solution of BG equation (1) more difficult. Because this equation represents a partial sum of infinitely many diagrams, we shall treat it with due care in section three.

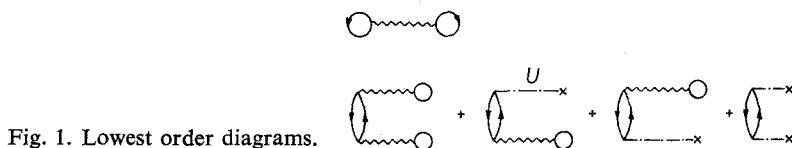


Fig. 1. Lowest order diagrams.

After obtaining the matrix elements of G , it is possible to evaluate the contributions of diagrams. We restrict ourselves to those depicted in Fig. 1 and write

$$(2) \quad E = E^{(1)} + E^{(2)},$$

$$E^{(1)} = f_1(A) \hbar\Omega + f_2(A) M(0, A),$$

$$\begin{aligned}
 E^{(2)} &= - \sum_{n=1}^{n_{\max}} 4^{-n} f_4(A) [M(n, A) - \Delta(n, A)]^2 / (\varepsilon_n - \varepsilon), \\
 M(n, A) &= \langle n000 \dots | G(2\varepsilon) | 0000 \dots \rangle + \\
 &\quad + \langle n0001 \dots | G(2\varepsilon) | 00001 \dots \rangle, \\
 \Delta(n, A) &= \delta_{n1} f_5(A) \hbar\Omega. \\
 \varepsilon &= 0.75 \hbar\Omega + f_3(A) M(0, A).
 \end{aligned}$$

The details are discussed thoroughly by Baranger [1], Afnan et al. [7], Sotona [3] and Blank [9]: The energy of the centre-of-mass motion is already subtracted in the approximation $T_{\text{cms}} = 0.75 \hbar\Omega$ and the matrix elements of the reaction matrix are considered in the standard double harmonic oscillator basis $|n1NLS \dots\rangle$ [1]. The additional quantum numbers $S \dots$ represent the spin etc. of the correlated nucleonic pair, single oscillator energy is $\varepsilon_n = \hbar\Omega(2n + l + 3/2)$. The symbol A denotes the number of nucleons in the nucleus and factors $f_1(3), \dots, f_5(3)$ and $f_1(4), \dots, f_5(4)$ are equal to $3/2, 3/2, 1, 3, (3/2)^{1/2}$, and $9/4, 3, 3/2, 9, (2/3)^{1/2}$, respectively.

Set (2) of equations is nonlinear since $\varepsilon \sim M(0, A)$ and $M(0, A)$ depends on ε . The solving will be performed by graphical interpolation. The convergence of the iterative procedure generating the mesh $\varepsilon = \varepsilon_i, i = 1, 2, \dots, N_\varepsilon$ in succession proved to be very quick. In case of the worst initialization, solution of (1) should be repeated seven times. The quick convergence leads also to the truncation of the infinite sum $E^{(2)}$ after $n_{\max} = 3$ terms. The detailed tables are omitted. It is seen that the only nontrivial operation is the solution of the BG equation (1).

3. SOLVING THE BG EQUATION

The two-step reference spectrum method we use was described and tested by Sauer [11]. Inclusion of the angle-averaged Pauli projector Q is done by the reference equation

$$(3) \quad G = G_r - G_r(\mathfrak{A} - \mathfrak{A}_r)G, \quad \mathfrak{A}_r = (\omega + T)^{-1}$$

solved by matrix inversion in a truncated LHO basis. The input reference matrix G_r is obtained from its definition

$$(4) \quad G_r = V - V\mathfrak{A}_r G_r$$

by the inversion on a mesh of grid points. Here, we add a few comments.

3.1. Approximations in reference equation

3.1.1. Q -averaging

The change of G caused by improving Q was tested in [12]. In the present context, qualitatively the same error estimates are obtained by artificially increasing or de-

creasing the matrix elements of the angle-averaged Q . In general, the change of the lowest matrix elements of G lies below 1% even for a very strong variation in Q . As a consequence of the non-projector approximation of Q , this influence is more pronounced when the numerator of $\mathfrak{U}$ is altered (cf. also [9]).

3.1.2. Basis truncation

The more serious ($>1\%$) error in G follows from exclusion of the nondiagonal elements of G_r in cms (LHO) quantum numbers. This is similar to our exact calculation [12] though the meaning of the diagonal approximation is somewhat different here due to the angle-averaging in Q . In simple tests, the drastic but nondiagonal cms cutoff $N \leq 3$, $n \leq 4$ proved to give better results than the standard $N = 0$, $n \leq 15$ one [11] using the comparable size of the matrices. In practice, the only complication connected with the necessity to evaluate the cms-nondiagonal LHO matrix elements of G_r will be discussed in the following subsection.

3.2. Approximations in reference matrix

3.2.1. Cms kinetic energy

On the basis of the numerical experience, Kahana et al. [13] have modified the standard approximation

$$(5) \quad T = k^2 + \sum_{N,L} |NL\rangle e_{NL}^S \langle NL|, \quad e_{NL}^S = \langle NL| K^2 |NL\rangle.$$

$$\mathbf{k} = (\mathbf{k}_1 - \mathbf{k}_2)/\sqrt{2}, \quad \mathbf{K} = (\mathbf{k}_1 + \mathbf{k}_2)/\sqrt{2}$$

(where $|NL\rangle$ denotes single LHO function of the cms momentum $\mathbf{K}$) by the prescription $e_{NL}^S \rightarrow e_{NL}^{\text{KLS}} = 0.8e_{NL}^S$. Starting with the exact continued fraction expansion [14] for the propagator

$$(6) \quad \mathfrak{U}_r = \frac{1}{\omega + k^2 + K^2} = \sum_{N,N',L} |NL\rangle \langle N'L| \frac{1}{\omega + k^2 + f_{NN'L}(k^2 + \omega)}$$

we may get exact values of the effective cms kinetic energy $f_{NN'L}(k^2 + \omega)$ (Table 1). We see that $f \sim e_{NL}^S$ approximation is valid for $k \rightarrow \infty$ only, while e_{NL}^{KLS} represents some mean value approximation for the integral over k . It is easy to improve this approximation by introducing the k -dependent factor $f^A \doteq f$ into denominator in (6). A few elementary approximations (fits) to the exact values f are shown in Table 2.

The most important consequence of better approximating $\mathfrak{U}_r$ is the possibility to include the nondiagonal cms couplings in (4) and calculate the cms-nondiagonal G_r . This is significantly simplified by the band structure of the resulting form of (4) since, according to Table 1, the cutoff of the form $|N - N'| \leq 1 \div 3$ should be used. One of the possible methods of solving such a "band-truncated" matrix equation is described in detail in [15].

We should mention here that the Tables 1 and 2 are designed for the kernel of (4) only. In (3), any improvements practically cancel and may be neglected [11].

Table 1
Effective cms kinetic energy $f_{NN'L}(\xi)$

N	N'	L	e_{NL}^S	$\xi = 1000$	$\xi = 50$	$\xi = 5$	$\xi = 1$
0	0	0	1.5	1.4985	1.4719	1.3119	1.0650
1	1	0	3.5	3.4935	3.3805	2.7457	1.8485
2	2	0	5.5	5.4846	5.2229	3.9463	2.4729
0	0	1	2.5		2.4540	2.2170	1.9084
1	1	1	4.5		4.3279	3.4910	2.5117
2	2	1	6.5		6.1392	4.5969	3.0320
0	0	2	3.5	3.4965		3.1390	2.8100
1	1	2	5.5	5.4876		4.2837	3.2946
0	1	0	∞	8.2×10^5	2.3×10^3	46.108	6.8144
0	2	0		3.7×10^8	5.5×10^4	170.59	11.356
0	3	0		1×10^{11}	9.8×10^5	604.50	23.960
1	2	0			—1370.5		—7.0573
1	3	0			2.3×10^4		10.248
2	3	0			—1024.7		—7.4490
0	1	1			1.8×10^4	45.276	8.7734
0	1	2		5.4×10^5		47.183	11.330

Table 2
Examples of approximate cms kinetic energy

N	N'	L	$f_{NN'L}^A(\xi)$	A	B	C
0	0	0	$A + B\xi$	1.1911	0.01817	
1	1	0		2.2864	0.06970	
2	2	0		3.1281	0.1275	
0	0	1		2.0528	0.02517	
0	0	0	$A - B/\xi$	1.3927	0.3433	
1	1	0		3.0481	1.2626	
2	2	0		4.4845	2.1351	
0	0	1		2.3243	0.4395	
0	0	0	$A - B/(\xi + C)^{1/2}$	1.5543	0.5517	0.2500
1	1	0		3.6476	2.0380	0.2500
0	0	1		2.5346	0.7123	0.2500
0	0	2		3.4957	0.7865	0.2500
1	2	0	$A + B\xi^C$	—5.918	—1.228	1.793
1	3	0		1.602	8.659	1.885
2	3	0		—6.076	—1.197	1.725

3.2.2. Grid points and interpolation formula

We have found that because of rounding in the computer, local potentials of the Reid type are better approximated by their separable equivalents than by the grid point mesh with a high number N_b of points. In the double precision and for $N_b = 16$, the matrix V becomes singular, i.e. the information may be compressed in the matrix of a lower dimension.

To prevent the growth of dimensions N_b , we have minimized them by choosing two independent types of the mesh $k = k_i$, $i = 1, 2, \dots, N_b$ for different types of integrands. In detail, we use the Gaussian points symbolized by the approximate unit projector I_G for inversion in the matrix equation (4). Then, the iterated form of (4)

$$(7) \quad \langle nl | G_r | ml' \rangle = \langle nl | I_L V I_L | ml' \rangle - \langle nl | I_L V I_G \mathfrak{U}_r I_G V I_L | ml' \rangle + \\ + \langle nl | I_L V I_G \mathfrak{U}_r G_r \mathfrak{U}_r I_G V I_L | ml' \rangle$$

is employed as the interpolation formula for projection of G_r on LHO functions with the Laguerre points I_L reflecting the exponential behaviour of LHO functions at infinity. Both convergences $I_G \rightarrow 1$ and $I_L \rightarrow 1$ are achieved and tested separately. In fact, the iteration in (7) is not necessary, it only allows to preserve the symmetry of matrices.

3.2.3. Grid points in the coordinate representation

For the more complicated local potentials, obtaining the k -representation of $V(r)$ is difficult. In this case the double numerical Fourier-Bessel transformation introducing additional numerical errors may be avoided when working with the r -space representation of the kernel $\mathfrak{U}_r$ instead of the k -space representation of $V(r)$. Since,

$$(8) \quad \langle rl | \frac{1}{\omega + k^2 + K^2} | r'l' \rangle = \delta_{ll'} \frac{1}{\sqrt{(rr')}} I_{l+1/2}(\gamma r_2) K_{l+1/2}(\gamma r_2), \\ r_1 = \min(r, r'), \quad r_2 = \max(r, r'), \quad \gamma^2 = K^2 + \omega$$

where $I_{l+1/2}$ and $K_{l+1/2}$ are the Bessel functions [16], (4) may be represented in the form

$$(9) \quad \langle rl' | G_r | nl \rangle = \sum_{l''} V_{l'l''}(r) [\delta_{l''l} \langle r | nl \rangle - \beta_{l''}(r) a_{l''m}(r) - \alpha_{l''}(r) b_{l''m}(r)], \\ a_{l''m}(r) = \int_0^r s^2 ds \alpha_{l''}(rs) \langle sl'' | G_r | nl \rangle, \\ b_{l''m}(r) = \int_r^\infty s^2 ds \beta_{l''}(rs) \langle sl'' | G_r | nl \rangle, \\ \alpha_l(r) = (\pi/2r)^{1/2} I_{l+1/2}(\gamma r), \\ \beta_l(r) = (2/\pi r)^{1/2} K_{l+1/2}(\gamma r).$$

The number of integrations remains the same, the behaviour of the integrands is sufficiently smooth and the convergence with respect to N_b is quick. This method may be considered a good alternative to the standard (differential) r -space methods described e.g. by Baranger [1].

4. THE DISCUSSION OF NUMERICAL RESULTS

The method described in section 2 determines the $\Omega^{(B)}$ parameter (i.e. H_0) from the condition of the uniform convergence of the perturbation expansion $E = E^{(1)} + E^{(2)} + \dots$. It is therefore of primary interest to compare the values of $\Omega^{(B)}$ and E with the results of the older calculations [7, 3] which perform the minimization of $E^{(2)}$. A priori, we could expect the worse value of E now because we do not use the semi-phenomenological C -shift parameter. It is a merit of $\mathfrak{U} = Q(\omega + QTQ)^{-1}Q$ propagator in (1) that the agreement achieved with present nonparametric approach is very good. It is shown in Table 3. We may conclude that the C -parameter is successfully eliminated since the difference between old and new E is mostly of the same order as the third order contribution, in spite of the change in G and Ω .

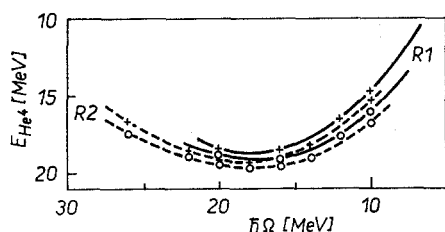


Fig. 2. First (+) and second (o) order He^4 energy for Reid soft core (R1) and modified (R2) potentials [21] and its dependence on $\hbar\Omega$.

Table 3

Ground state energy compared with $C \neq 0$ results of Afnan et al. [7] (first 3 columns) and Sotona [3]

Nucleus, $A =$ Potential	4 PD1 ^{a)}	4 PD4 ^{a)}	4 PD7 ^{a)}	4 NGL ^{b)}	3 M2 ^{c)}
Energy per nucleon E/A [MeV]	9.76	7.32	5.66	10.62	3.54
Third order contribution	0.26	0.41	0.60		
LHO frequency $\hbar\Omega$	34.0	26.0	22.0	34.27	18.4
Our result E/A	9.58	7.04	4.89	10.52	3.54
$\hbar\Omega^{(B)}$	35.89	28.26	21.54	44.3	24.32
Deviation in energy/ A	0.18	0.28	0.77	0.10	0.00

^{a)} Afnan et al. [7]; ^{b)} Narodetskii [19]; ^{c)} Mongan [20].

The knowledge of exact value of $\Omega^{(B)}$ (size of LHO well) is less important since the $E(\Omega)$ dependence has a broad minimum (Fig. 2). Nevertheless, the correct values of E for different potentials should be stable and related to the admixture of the D-state probability (PD) in deuteron. This dependence (Coester line) is well reproduced by our results (Fig. 3) for both local and separable realistic potentials. This confirms the internal consistency of the Baranger's prescription for Ω .

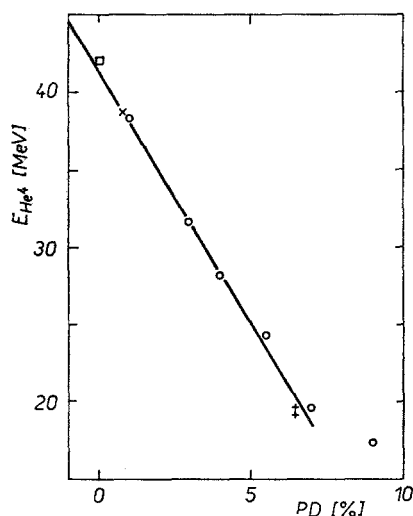


Fig. 3. Correlation between PD and E in He^4 for realistic potentials PD1 — PD7 ([7], $\circ$), M2 ([20], $\times$), R1, R2 ([21], $+$) and NGL ([19], $\square$).

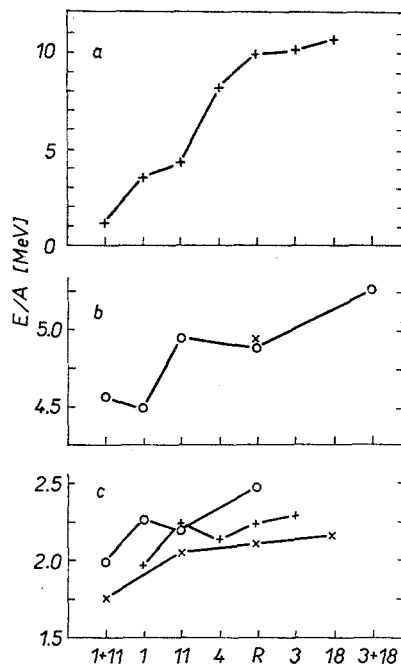


Fig. 4. Off-shell influence for ground-state energy in H^3 (c), He^4 (b), and nuclear matter (a). Comparison with the results of Haftel and Tabakin [17] (a) and Kim et al. [4] ($+$, $\times$) for different phase-equivalent transforms (1 + 11, ..., cf. [4]) of the Reid potential [21].

Our main interest concerns the comparison of E with "exact" results. After all, the recent appearance of the solutions [5] of Faddeev-Yakubovsky equations for the Reid potential was our main motivation to continue the comparative test of Sotona. The results are presented in Table 4. The deviations are of the same magnitude as in Table 3.

Of course, the main attention is paid to the Reid potential. To avoid the situation of comparing just two numbers, we employ the whole set of potentials related by the phase-equivalent transformations. We may employ the existing calculations of the nuclear matter (Haftel and Tabakin [17], Fig. 4a) and also the Faddeev equa-

tion solutions (Kim et al. [4], Fig. 4c). Ordering the set of potentials according to the increasing binding of the nuclear matter, we obtain the results (o) depicted in Fig. 4b, c. The off-shell dependence is roughly monotonous again, the errors of both BT and Faddeev equation seem to be comparable and all the values of E for H^3 to be mutually consistent. Thus, the off-shell influence is well reproduced, in spite of the rough nature of BT approximation.

Table 4
Ground-state energies compared with the nonperturbative results.

Nucleus	E^{exact} [MeV]	Reference	Potential	E^{perturb}	Δ/A
H^3	10.16	[3]	M2	10.61	0.15
	6.71	[4] ^{a)}	R1	7.44	0.24
	6.37	[4] ^{b)}	R1	7.44	0.37
	5.28	[4] ^{b)}	R1, FE 1 + 11 ^{c)}	5.99	0.24
	5.92	[4] ^{a)}	R1, FE 1 ^{c)}	6.81	0.30
	6.70	[4] ^{a)}	R1, FE 11 ^{a)}	6.64	-0.02
	6.19	[4] ^{b)}	R1, FE 11 ^{c)}	6.64	0.15
He^4	45.73	[3], [19]	NGL	42.09	0.91
	19.5	[5]	R1	19.18	-0.08

^{a)} two-component solution of Faddeev equations

^{b)} three-component solution

^{c)} different phase-equivalent transformations of potential as introduced in [4]

There is vast literature on improving the precision of the low-order approximation (see e.g. the references in [1]). It seems to us that from the mathematical point of view, the next step should not be the higher-order contribution evaluation but rather the computer-oriented calculation of an "exact" (variational) result. From the physical point of view, it is a question whether we should further improve the simple methodology at all because the present error level is not entirely incomparable with the very limitations of the potential model ignoring the mesonic degrees of freedom (cf. e.g. the discussion in Kümmel et al. [2]).

5. CONCLUSIONS

It remains to compare the motivations and results of the present paper in the light of the work and results achieved in the exp S theory [2]. We may conclude that

(i) The rate of convergence of perturbation expansion differs for different quantities. Concerning the belief in the applicability of the low order BT for the approximate

ground-state energy calculation in the finite nuclei, we have supported it by the direct comparison with the "exact" results of the Faddeev and Faddeev-Yakubovsky integral equations. No free parameters influence the results and the test is performed for the whole classes of phase-equivalent potentials so that the reliability of the method and Sotona's conclusions are further confirmed.

(ii) The results are practically given by solving one (BG) equation. Nevertheless, they roughly reproduce even the off-shell differences between potentials. This may encourage the introduction of G and elementary correlations even in some non-BT applications.

(iii) After all, any exact computer solution of high complexity should be interpreted in a physical language tracing the important influence and suppressing the irrelevant factors. The study of the cms or Pauli principle influence in BG equation is an example of such an analysis which is possible even when using quite crude approximations.

(iv) The methodical part of the paper is of interest in itself. We have discussed the improvements of the Sauer's method. We expect that this will be interesting especially in the heavier nuclei where higher LHO quantum numbers play a role. Possibly, the questions of precision of the approximate solution of BG-like equations may also be of interest in $\exp S$ context — cf. e.g. the effect of technical improvements on the photonuclear enhancement factor calculations (Gari et al. [18]).

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